

The Sommerfeld constant and the exact 66-degree angle carrier

Electromagnetic spectral invariance, a determinant threshold sum rule and the gauge boundary

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An exact golden-coefficient identity and a continuous weak-phase orbit at fixed one-loop photon screening. Explicit separation of electromagnetic coupling inputs, scalar angles and physical CP.

The constants and angles being compared

Sommerfeld's fine-structure constant is the electromagnetic coupling α , not the strong-CP determinant phase or the unitarity-triangle angle also conventionally called α . In SI units $\alpha = e^2/(4 \pi \epsilon_0 \hbar c)$. The latest available CODATA adjustment is 2022: $\alpha(0) = 0.0072973525643$ with standard uncertainty 0.000000000011, and $\alpha(0)$ -inverse = 137.035999177 with standard uncertainty 0.000000021. These are measured references, not inputs used to calibrate the new phase-orbit calculation. NIST's current constants table and the PDG 2026 electroweak review retain these values. The coupling at nonzero momentum is scale dependent; a proposed exact formula must specify the physical limit, renormalization convention and threshold matching.

The nominated scalar angle $\theta = 66$ degrees is $11 \pi/30$ radians. It is distinct from the standard CKM phase δ , the unitarity-triangle angles, and the strong-CP phase θ -bar. The new Hermitian action protects θ -bar through first-order one-loop order while allowing weak CP. That protection does not identify its scalar source frame with an independently locked 66-degree field. An equation connecting those fields to quark matrices is still necessary. Comparing decimal constants or using the degree number 66 inside a coupling formula would not supply that equation; physically meaningful angle formulas must survive changing angle units.

The exact 66-degree algebraic relation

$$\theta = 11\pi/30, \quad 1 - 2\cos(12\theta) = (3 - \sqrt{5})/2$$

For $z = \exp(11 \pi i/30)$, z has primitive order sixty. Its cyclotomic polynomial is $\Phi_{60}(z) = z^{\text{to-the-sixteenth}} + z^{\text{to-the-fourteenth}} - z^{\text{to-the-tenth}} - z^{\text{to-the-eighth}} - z^{\text{to-the-sixth}} + z^{\text{to-the-fourth}} + 1$. Set $x = z + z^{-1} = 2 \cos(\theta)$. The exact real minimal polynomial is $P(x) = x^{\text{to-the-eighth}} - 7x^{\text{to-the-sixth}} + 14x^{\text{to-the-fourth}} - 8x^{\text{to-the-second}} + 1$, with $\Phi_{60}(z) = z^{\text{to-the-eighth}} P(z + z^{-1})$. The receipt verifies this polynomial identity symbolically and independently reuses the rational quotient-algebra carrier.

The twelfth power maps the selected root to $\exp(2 \pi i/5)$, so $C = 1 - z^{\text{to-the-twelfth}} - z^{\text{to-the-minus-twelfth}} = 1 - 2 \cos(12 \theta) = (3 - \sqrt{5})/2 = \phi^{\text{to-the-minus-two}}$. Its exact minimal polynomial is $C^2 - 3C + 1 = 0$. This is a genuine algebraic connection between the nominated angle and golden coefficient. It contains no electromagnetic coupling.

Moreover the positive-half-circle primitive-sixtieth-root angles yielding the smaller coefficient are exactly 6, 66, 114 and 174 degrees. All have negative partners with the same CP-even coefficient. The coefficient therefore loses the embedding information needed to distinguish 66 degrees. Our previously certified six-harmonic potential does select plus/minus 66 degrees among competitors, but its selected coefficient relations were constructed with that target known. A successful joint mechanism must justify the energy selector and transmit its selected phase to normalized quark matrices, not only repeat this quotient identity.

A photon theorem for the protected canonical action

$$\sum_{i=1}^6 \log(\mu/m_i) = 6 \log \mu - 3 \log(v_{\text{ym}})$$

Each complete up sector contains six Dirac states of electric charge $+2/3$, and each down sector six of charge $-1/3$. The ordinary doublet and vectorlike singlet components have the same electric charge within a sector. Consequently both chiral photon generators are Q_f times the six-dimensional identity. Independent left and right fermion diagonalization leaves each generator unchanged. The one-loop fermion photon vacuum polarization therefore depends only on the six singular masses; it cannot see their mixing eigenframes.

At spacelike momentum magnitude Q define the positive inverse-alpha screening contribution $S_f(Q) = (2 N_c Q_f^2/\pi) \sum_i \int_0^1 dx x(1-x) \log[1+Q^2 x(1-x)/m_i^2]$, with $N_c = 3$. Then $\alpha_{\text{eff}}(Q)^{-1} = \alpha(0)^{-1} - \sum_f S_f(Q)$ at the stated one-loop accuracy, omitting other particle contributions. This expression is a perturbative free-fermion contribution in declared model units. It does not replace physical hadronic vacuum polarization, lepton and charged-gauge contributions, or electroweak matching required to relate the model to measured $\alpha(0)$. The numerical integral independently passes the small- Q normalization $S_f = N_c Q_f^2 Q^2/(15 \pi m^2)$ for one mass, as well as scale covariance and zero-momentum checks.

An even stronger logarithmic identity follows from the triangular determinant. For $D_f = [[v y_f I, 0], [C_f, m_f I]]$, the product of all six positive singular masses is $|\det D_f| = (v y_f m_f)^3$, independent of C_f . Hence $\sum_i \log(\mu/m_i) = 6 \log(\mu) - 3 \log(v y_f m_f)$. The complete one-loop logarithmic threshold, $(2 N_c Q_f^2/(3 \pi))$ times this sum in the declared integration convention, is source-independent even when C_f changes its eigenvalues. This exact statement requires retaining both light and heavy partners. Heavy-only thresholds can depend on the light masses; their dependence cancels against the light-partner contribution in the all-state logarithmic sum. The all-state logarithmic result does not imply that a finite-momentum polarization is independent of varying masses.

A continuous weak-phase orbit at fixed electromagnetic screening

Retain the preceding canonical action, illustrative six light masses, $v = 0.03$, $y_{\text{up}} = 1.2$, $y_{\text{down}} = 0.57$ and universal vectorlike masses one. Fix C_{up} diagonal and construct $C_{\text{down}} = U(\delta) \text{diag}(h_i) U(\delta)^\dagger$, with the exact finite-mass inverse $h_i^2 = (a^2 - \ell_i^2)(m^2 - \ell_i^2)/\ell_i^2$. $U(\delta)$ is a standard unitary three-angle chart with fixed $\sin(\theta_{12}) = 0.22517$, $\sin(\theta_{13}) = 0.003763$ and $\sin(\theta_{23}) = 0.04189$. Its δ is varied continuously, not predicted. This is a counterexample family within the allowed source-center coefficient space.

At every point all six light masses, all six heavy partner masses, and all doublet weights remain fixed. The exact source-center covariant determinant permits each new Hermitian center to be reconstructed by nine real polynomial coefficients at the retained scalar vacuum. The squared-lock scalar EFT then keeps that vacuum stationary and locally positive by the previous congruence proof. The tree and one-loop strong-CP mass-phase theorem remains applicable. Thus the variation changes justified scalar-center inputs without changing the complete five-coefficient fermion action or introducing a new quark metric.

The receipt samples $\delta = 0, 6, 42, 60, 66, 78, 90, 114, 138, 174, 180$ and -66 degrees. It records all center coefficients, recovered unitary charts after removing exact finite doublet weights, finite-current CP quartets, complete spectra, and photon integrals at $Q = 10^{-5}, 0.01, 1$ and 100 in model units. The largest relative mass change is 3.78×10^{-15} . The largest screening change is 8.89×10^{-16} in inverse-alpha units, and the maximum all-state logarithmic identity error is 4.45×10^{-16} . The analytic fixed-spectrum identity, rather than those numerical residuals,

establishes invariance for the whole continuous orbit. Weak CP changes along that orbit, including its sign at the conjugate branch.

The independent gauge boundary and the mechanism requirement

$$\mathcal{L}_Y = -\frac{k_0}{4} F_{\mu\nu} F^{\mu\nu}, \quad \alpha^{-1} = 4\pi k_0$$

In the unit-charge covariant-derivative normalization the gauge action contains $-k_0 F_{\mu\nu} F^{\mu\nu}/4$, where $k_0 = 1/e^2$ and $\alpha^{-1} = 4\pi k_0$. This operator is invariant under every declared finite flavor group, shaping charge and CP transformation. Its positive coefficient is independent of the scalar source-center coefficients. Equivalently, above electroweak breaking, $\alpha^{-1} = 4\pi (g_Y^{-2} + g_2^{-2})$ at tree matching. The finite flavor symmetry does not normalize either continuous gauge coupling.

A permitted scalar-dependent gauge kinetic interaction can create a dependence on scalar invariants, but it retains the independent constant k_0 . Specifying its coefficient to make 137.035999177 emerge would match the observed coupling, not predict it. Conversely inserting α into the source potential or fixing a source angle to 66 degrees would initialize the desired correlation. The new result precisely rules out using equal-charge one-loop mass thresholds as the missing frame-selection mechanism: complete photon screening remains unchanged along an allowed continuous weak-phase orbit, and its all-state logarithmic threshold is even independent of source eigenvalues.

A substantive extension must supply a justified UV gauge normalization or boundary condition, an interaction that fixes the source-center freedom, an embedding selector for the primitive-root angle, and their physical matching. A common scalar modulus alone supplies none of these coefficient relations automatically. Higher-loop weak/flavor corrections can carry mixing dependence and are not excluded by the one-loop theorem. They also require specified gauge and scalar inputs before they can predict a number. The current action therefore provides an exact golden-angle carrier and exact electromagnetic matching sum rules, not a derivation of the Sommerfeld constant from that angle.

Reproduction and references

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_electromagnetic.py` and `PYTHONPATH=python python python/tests/test_flavor_electromagnetic.py`. All six new and eighteen related checks pass; fresh replay reproduces the receipt byte for byte. Both PDFs retain the calculation in full.

Primary references: NIST, 2022 CODATA constants table; PDG, Electroweak Model and Constraints on New Physics (2026), equations 10.19–10.20. Their primary URLs are retained in the receipt. The determinant and source-frame statements follow directly from the specified action. Standard QED screening and gauge-normalization freedom are established physics.